Mujeeb Rahman P M

Software Engineer (M.Sc.) | C++, C#, Unity, Debugging and Multi-Platform Builds

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PROFILE

Software engineer (M.Sc.) working in both C++ and C#. At Finastra I was the sole developer of a C++ module in production for eighteen months: reproducing reported defects, tracing them to their cause in a codebase I had not written, and verifying that existing behaviour was unchanged before each release. At Sartorius I spent eighteen months building and shipping Unity applications across HoloLens 2, iPad Pro, Android tablets and Apple Vision Pro — including the device-side debugging and performance work that platform builds require. My master's thesis at Fraunhofer IPM involved a native C++ module driven from a managed engine runtime across a DLL boundary. English C2.

EXPERIENCE

C++/Qt Application Development | Freelance | Villingen-Schwenningen

June 2025 to present

- Building a C++/Qt desktop application end to end — architecture, interface, data layer, testing and documentation.
- Developed and tested on both Linux and Windows; CMake build, Git version control.

Technologies: C++, Qt Widgets, SQLite, CMake, Git, Linux, Windows

Master's Thesis: Real-Time Point Cloud Rendering | Fraunhofer IPM | Freiburg

Sept. 2024 to May 2025

- Implemented a native C++ module called from a managed engine runtime (Godot) across a DLL boundary — marshalling point data each frame and returning the visible set for rendering.
- Worked into an unfamiliar open-source codebase from another institute: analysed its structure and dependencies, then restructured the host side from static file input to continuous streaming while leaving the core algorithms intact.
- Diagnosed a throughput limitation by measuring where the constraint actually was rather than assuming, then implemented a GPU-based replacement for the limiting stage — capacity rose from ~1.8M to ~300M points per second and was adopted by the project team.
- Wrote a Python tool reproducing the sensor's output format and timing, which allowed development and regression testing without the hardware connected.

Technologies: C++, CUDA, Python, ROS, C#, Godot, OpenXR, CMake, Linux, Git, Jira, Confluence

XR Application Development (Internship) | Sartorius | Göttingen

Oct. 2023 to March 2024

- Built a Unity application for HoloLens 2 in a three-person team, guiding operators through multi-step procedures with content anchored to equipment via markers.
- Profiled and optimised rendering and application performance for standalone headset hardware; debugged directly on device, where behaviour differed from the editor.
- Designed and implemented the full user interface, including an onboarding module for first-time headset users; coordinated reviews with the UX and brand teams.
- Commissioned and validated the build on site at customer laboratories, through to formal acceptance.

Technologies: Unity, C#, MRTK3, OpenXR, HoloLens 2, Git, Jira

XR Software Development (Working Student) | Sartorius | Göttingen

Oct. 2022 to Sept. 2023

- Built and shipped Unity applications across several target platforms — HoloLens 2 (UWP), iPad Pro (iOS) and HP Reverb — handling the per-platform build configuration and device testing each required.
- Developed a visionOS prototype in Xcode and Reality Composer Pro; selected with this project for an Apple Developer Lab in Munich and worked with Apple Vision Pro hardware before market launch.
- Built a multi-user prototype allowing several people in different locations to share one scene: networking over Unity Netcode and Unity Gaming Services, session join by ID, avatars and hand tracking.
- Prepared and optimised CAD-derived geometry for real-time rendering on constrained hardware using Pixyz and Blender — mesh decimation and retopology.

Technologies: Unity (UI Toolkit, Canvas), C#, MRTK, Unity Netcode, Xcode, Blender, Pixyz, HoloLens 2, iPad Pro, Apple Vision Pro, Git, Jira

UI Development (Working Student) | TeamViewer | Göppingen Sept. 2021 to Feb. 2022

- Developed and integrated UI components against an existing C++ backend for a Windows desktop application, working into the application's existing structures.
- Functional testing and defect analysis before release.

Technologies: C++, Win32 API, JavaScript, HTML, CSS, Git, Jira

Associate Software Engineer | Finastra | Trivandrum, India Sept. 2019 to March 2021

- Sole developer responsible for a C++ module inside a production system — maintaining a codebase that had grown over years and that I had not written.
- Investigated defects reported through the ticket system: reproduced the reported behaviour, traced it to its cause and implemented the fix — including defects that only appeared under particular customer configurations.
- Ran regression checks before every release to confirm existing functionality was unaffected; ran static analysis (Coverity, later Checkmarx) and remediated findings.
- Technical contact for migrating the application to a newer Visual Studio version and platform toolset, including handover sessions for an external team.
- Also built XML-based document templates for a C#/WPF application; worked across two parallel Scrum teams with colleagues at the company's US site.

Technologies: C++, MFC, C#, WPF, Visual Studio, Coverity, Checkmarx, Azure DevOps, Git, Jira, Confluence

PERSONAL PROJECTS

Unity simulation of an automated guided vehicle streaming camera output and telemetry to a second device — WebRTC for video, WebSocket for telemetry, with on-screen latency measurement.

Game prototype in Unreal Engine 5.6 (C++ and Blueprints), ported from an earlier Unity version; low-poly environment assets authored in Blender.

Android AR prototype displaying segmented medical imaging data, with layer visibility, transparency and a clipping plane (3D Slicer, Blender, Unity).

TECHNICAL SKILLS

Languages: C++ (C++14/17/20), C#/.NET, Python, JavaScript

Unity: Editor, UI Toolkit, Canvas, MRTK/MRTK3, OpenXR, Netcode for GameObjects; builds for Windows/UWP, iOS, Android and visionOS

Debugging & Quality: Reproducing reported defects, root-cause analysis in unfamiliar code, on-device debugging, regression testing, static analysis (Coverity, Checkmarx), code reviews

Graphics: Real-time rendering, GPU programming (CUDA), native/managed interop across a DLL boundary, point cloud processing, mesh decimation and retopology

Platforms: Windows, Linux, HoloLens 2, iPad Pro, Android tablets, Apple Vision Pro, HP Reverb

Tools: Git, CMake, Visual Studio, Xcode, CI/CD (Azure DevOps), Jira, Confluence

EDUCATION

M.Sc. Global Software Development — Hochschule Fulda, Germany | 2021 to 2025

B.Tech. Computer Science and Engineering — APJ Abdul Kalam Technological University, India | 2015 to 2019

LANGUAGES

English: C2 (fluent) | German: B2